

FitsAI

Multi-Agent AI System Implementation

Central Bank of Oman

Technical Architecture & Delivery Roadmap

WEEKS 1-2

Foundation

WEEKS 3-4

Frontend

WEEKS 5-6

Content Gen

WEEKS 7-8

Integration

Phase 1: Foundation & Core Agents (Weeks 1-2)

Week 1: Infrastructure & Basic AI Setup

Day 1-3: Deploy n8n, PostgreSQL, ChromaDB on GCP. Configure AI APIs (Vertex, Gemini, OpenAI). Setup JWT auth.

Day 4-7: Build core n8n workflow for document processing (Upload → OCR → Embeddings → Storage). Implement basic vector search.

Week 2: Multi-Agent Core

AI Agents: Implement Knowledge, IT Support, and Content Generator agents. Create n8n router workflow.

Basic Content Gen: Setup jsPDF for PDFs, DALL-E 3 for infographics, and D-ID API for simple avatar videos.

Phase 2: Frontend Development (Weeks 3-4)

Week 3: Core Frontend Architecture

Setup React.js + Material-UI. Implement authentication, role-based routing, base layout, universal chat interface, and WebSocket integration.

Week 4: Role-Specific Dashboards

Parallel development of four dashboards: **Employee** (chat, search), **Admin** (monitoring, user mgmt), **Support** (tickets, metrics), and **Editor** (approvals, templates).

Phase 3: Content Generation & Backend APIs (Weeks 5-6)

Week 5: Enhanced Content Generation

Develop robust pipelines for branded PDFs (jsPDF), styled infographics (DALL-E 3 + Canva), and avatar videos with multi-language voiceovers (D-ID + OpenAI TTS).

Week 6: Backend API Development

Build Node.js/Express API endpoints for auth, document CRUD, content requests, real-time chat, file handling, and ITMS integration.

Phase 4: Integration & Production (Weeks 7-8)

Week 7: ITMS Integration & Advanced Features

Connect to CBO ITMS via REST API for ticket sync. Implement feedback collection, content queues, multi-language support, and advanced search filters.

Week 8: Production Deployment

Deploy to GCP production, configure load balancing, harden security, optimize performance, and set up basic monitoring for system handover.

Condensed Technical Stack

Orchestration: n8n

AI: Vertex AI, Gemini, GPT-4, DALL-E 3

Database: PostgreSQL, ChromaDB

Backend: Node.js, Express, WebSockets

Frontend: React.js, Material-UI

Content Gen: jsPDF, Chart.js, D-ID, Canva

Cloud: Google Cloud Platform

Auth: JWT Tokens

Rapid n8n Workflows

- Universal Content Request**
User Input → Intent Detection → Agent Router → Content Generator → Output
- ITMS Sync**
Ticket Event → Data Transform → CBO ITMS API → Status Update
- Quick Media Generation (PDF/Video)**
Request → Template → Data Fetch → Assembly (jsPDF/D-ID) → Delivery

Streamlined Architecture

